



AI POWERED EV BATTERY SYSTEM

INTELLIGENT BATTERY HEALTH DASHBOARD

DARK

LIGHT

SOH STATUS

--%

CHARGE LEVEL

--%

TEMPERATURE

--°C

SYSTEM



STANDBY

SOH ESTIMATION

RL OPTIMIZATION

MODEL COMPARISON

STATE OF HEALTH ESTIMATION

LSTM Model



▶ START SYSTEM

UPLOAD BATTERY DATA

Click to select a single CSV file

PROGRESS

0 / 0

MODEL MAPE

2.62%

BATTERY STATUS

--



Predicted SOH (%)



Actual SOH (%)

1.0

0.9



AI POWERED EV BATTERY SYSTEM

INTELLIGENT BATTERY HEALTH DASHBOARD

DARK

LIGHT

SOH STATUS

--%

CHARGE LEVEL

--%

TEMPERATURE

--°C

SYSTEM



STANDBY

SOH ESTIMATION

RL OPTIMIZATION

MODEL COMPARISON

REINFORCEMENT LEARNING LOGS

Q-Learning (Baseline) ▼

RECOMMENDED ACTION

--

CUMULATIVE REWARD

0.0

ACTION EFFICIENCY

--%

Project: On-Time Battery Health Estimation and Optimization in Electric Vehicles

Team: Md. Shiban Alam, Syed Sarafeena Ali, Purbasha Roy, Dindira Sain

Mentor: Mrs. Amrita Bhattacharya | **Institution:** STCET - MAKAUT

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Click to select a single CSV file

PROGRESS

0 / 0

MODEL MAPE

2.62%

BATTERY STATUS

--

Predicted SOH (%) Actual SOH (%)

1.0

0.9

★ Quick access  This PC  3D Objects  Desktop  Documents  Downloads  Music  Pictures  Videos  Local Disk (C:)  Network	Name	Date modified	Type
	 nasa_test	01-05-2026 16:08	Microsoft Excel
	 nasa_train	01-05-2026 16:08	Microsoft Excel
	 nasa_val	01-05-2026 16:09	Microsoft Excel
	⏪ ⏩		

🌟 Quick access 💻 This PC 📦 3D Objects 🖥 Desktop 📄 Documents ⬇ Downloads 🎵 Music 🖼 Pictures 🎬 Videos 💾 Local Disk (C:) 🌐 Network	Name	Date modified	Type
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STANDBY

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STATE OF HEALTH ESTIMATION

LSTM Model



▶ START SYSTEM

UPLOAD BATTERY DATA

Click to select a single CSV file

Loaded: nasa_test.csv | Total Cycles Found: 73

PROGRESS

0 / 0

MODEL MAPE

2.62%

BATTERY STATUS

--

Predicted SOH (%) Actual SOH (%)



AI POWERED EV BATTERY SYSTEM

INTELLIGENT BATTERY HEALTH DASHBOARD

DARK

LIGHT

SOH STATUS

--%

CHARGE LEVEL

--%

TEMPERATURE

--°C

SYSTEM



STANDBY

SOH ESTIMATION

RL OPTIMIZATION

MODEL COMPARISON

STATE OF HEALTH ESTIMATION

LSTM Model



▶ START SYSTEM

UPLOAD BATTERY DATA

Click to select a single CSV file

Loaded: nasa_test.csv | Total Cycles Found: 73

PROGRESS

0 / 0

MODEL MAPE

2.62%

BATTERY STATUS

--

Predicted SOH (%) Actual SOH (%)



AI POWERED EV BATTERY SYSTEM

INTELLIGENT BATTERY HEALTH DASHBOARD

DARK

LIGHT

SOH STATUS

--%

CHARGE LEVEL

--%

TEMPERATURE

--°C

SYSTEM



ACTIVE

SOH ESTIMATION

RL OPTIMIZATION

MODEL COMPARISON

STATE OF HEALTH ESTIMATION

LSTM Model



STOP

UPLOAD BATTERY DATA

Click to select a single CSV file

Live SOH Estimation Started...

PROGRESS

0 / 0

MODEL MAPE

2.62%

BATTERY STATUS

--

Predicted SOH (%) Actual SOH (%)



AI POWERED EV BATTERY SYSTEM

INTELLIGENT BATTERY HEALTH DASHBOARD

DARK

LIGHT

SOH STATUS

72.0%

CHARGE LEVEL

72.8%

TEMPERATURE

36.8°C

SYSTEM



ACTIVE

SOH ESTIMATION

RL OPTIMIZATION

MODEL COMPARISON

STATE OF HEALTH ESTIMATION

LSTM Model



STOP

UPLOAD BATTERY DATA

Click to select a single CSV file

Live SOH Estimation Started...

PROGRESS

9 / 73

MODEL MAPE

2.62%

BATTERY STATUS

Degraded

Predicted SOH (%) Actual SOH (%)

80

PROGRESS

18 / 73

MODEL MAPE

2.62%

BATTERY STATUS

Degraded



Project: On-Time Battery Health Estimation and Optimization in Electric Vehicles

Team: Md. Shibon Alam, Syed Sarafeena Ali, Purbasha Roy, Dindrila Sain

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AI POWERED EV BATTERY SYSTEM

INTELLIGENT BATTERY HEALTH DASHBOARD

DARK

LIGHT

SOH STATUS

64.6%

CHARGE LEVEL

60.8%

TEMPERATURE

39.8°C

SYSTEM



ACTIVE



SOH ESTIMATION



RL OPTIMIZATION



MODEL COMPARISON

REINFORCEMENT LEARNING LOGS

Q-Learning (Baseline) ▼

RECOMMENDED ACTION

Moderate

CUMULATIVE REWARD

48.0

ACTION EFFICIENCY

85.0%

Project: On-Time Battery Health Estimation and Optimization in Electric Vehicles

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PROGRESS

28 / 73

MODEL MAPE

2.62%

BATTERY STATUS

Degraded



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AI POWERED EV BATTERY SYSTEM

INTELLIGENT BATTERY HEALTH DASHBOARD

DARK

LIGHT

SOH STATUS

62.9%

CHARGE LEVEL

51.2%

TEMPERATURE

42.2°C

SYSTEM



ACTIVE

SOH ESTIMATION

RL OPTIMIZATION

MODEL COMPARISON

STATE OF HEALTH ESTIMATION

LSTM Model



STOP

UPLOAD BATTERY DATA

Click to select a single CSV file

Live SOH Estimation Started...

PROGRESS

36 / 73

MODEL MAPE

2.62%

BATTERY STATUS

Degraded

Predicted SOH (%) Actual SOH (%)

80



AI POWERED EV BATTERY SYSTEM

INTELLIGENT BATTERY HEALTH DASHBOARD

DARK

LIGHT

SOH STATUS

61.9%

CHARGE LEVEL

47.2%

TEMPERATURE

43.2°C

SYSTEM



ACTIVE



SOH ESTIMATION



RL OPTIMIZATION



MODEL COMPARISON

REINFORCEMENT LEARNING LOGS

Q-Learning (Baseline)



RECOMMENDED ACTION

Moderate

CUMULATIVE REWARD

82.0

ACTION EFFICIENCY

85.0%

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PROGRESS

46 / 73

MODEL MAPE

2.62%

BATTERY STATUS

Degraded



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AI POWERED EV BATTERY SYSTEM

INTELLIGENT BATTERY HEALTH DASHBOARD

DARK

LIGHT

SOH STATUS

61.9%

CHARGE LEVEL

36.8%

TEMPERATURE

45.8°C

SYSTEM



ACTIVE

SOH ESTIMATION

RL OPTIMIZATION

MODEL COMPARISON

STATE OF HEALTH ESTIMATION

LSTM Model



STOP

UPLOAD BATTERY DATA

Click to select a single CSV file

Live SOH Estimation Started...

PROGRESS

54 / 73

MODEL MAPE

2.62%

BATTERY STATUS

Degraded

Predicted SOH (%) Actual SOH (%)

80

70



AI POWERED EV BATTERY SYSTEM

INTELLIGENT BATTERY HEALTH DASHBOARD

DARK

LIGHT

SOH STATUS

63.4%

CHARGE LEVEL

32.8%

TEMPERATURE

46.8°C

SYSTEM



ACTIVE



SOH ESTIMATION



RL OPTIMIZATION



MODEL COMPARISON

REINFORCEMENT LEARNING LOGS

Q-Learning (Baseline)



RECOMMENDED ACTION

Moderate

CUMULATIVE REWARD

118.0

ACTION EFFICIENCY

85.0%

Project: On-Time Battery Health Estimation and Optimization in Electric Vehicles

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PROGRESS

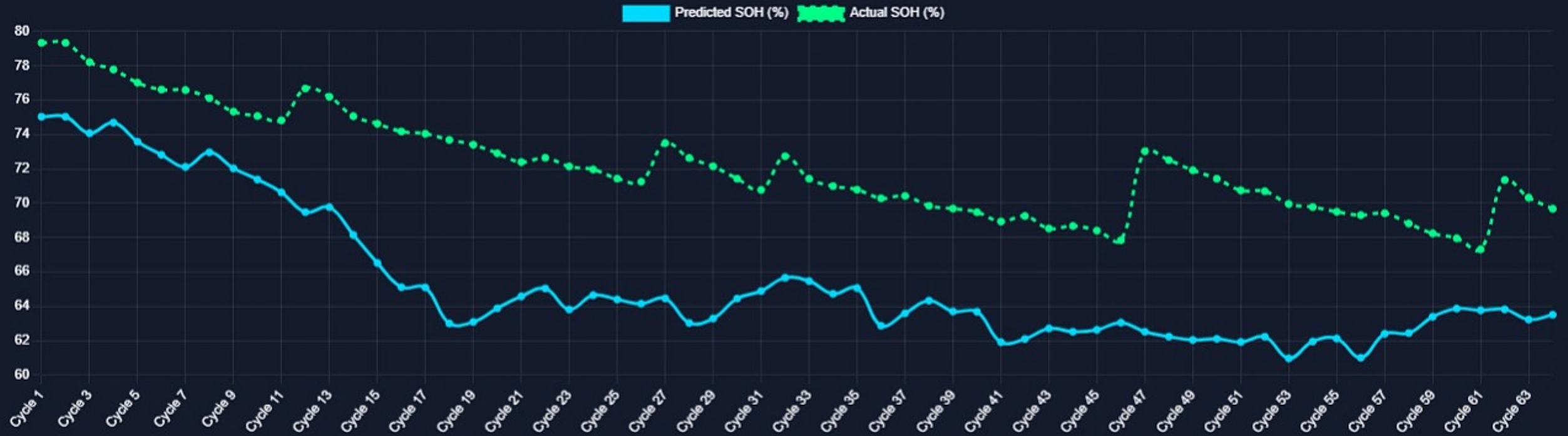
64 / 73

MODEL MAPE

2.62%

BATTERY STATUS

Degraded



Project: On-Time Battery Health Estimation and Optimization in Electric Vehicles

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AI POWERED EV BATTERY SYSTEM

INTELLIGENT BATTERY HEALTH DASHBOARD

DARK

LIGHT

SOH STATUS

63.2%

CHARGE LEVEL

21.6%

TEMPERATURE

49.6°C

SYSTEM



PROCESSED

SOH ESTIMATION

RL OPTIMIZATION

MODEL COMPARISON

STATE OF HEALTH ESTIMATION

LSTM Model



START SYSTEM

UPLOAD BATTERY DATA

Click to select a single CSV file

Route Finished: All Telemetry Processed.

PROGRESS

73 / 73

MODEL MAPE

2.62%

BATTERY STATUS

Degraded

Predicted SOH (%) Actual SOH (%)

80

70

PROGRESS

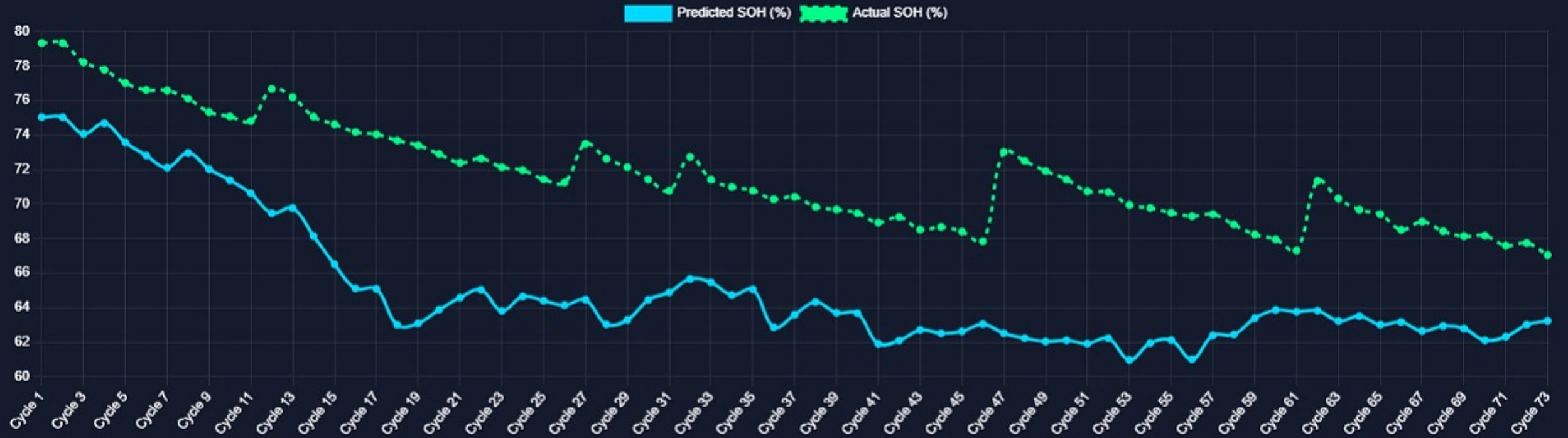
73 / 73

MODEL MAPE

2.62%

BATTERY STATUS

Degraded



Project: On-Time Battery Health Estimation and Optimization in Electric Vehicles

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LIGHT

SOH STATUS

63.2%

CHARGE LEVEL

21.6%

TEMPERATURE

49.6°C

SYSTEM



PROCESSED

SOH ESTIMATION

RL OPTIMIZATION

MODEL COMPARISON

REINFORCEMENT LEARNING LOGS

Q-Learning (Baseline) ▼

RECOMMENDED ACTION

Moderate

CUMULATIVE REWARD

146.0

ACTION EFFICIENCY

85.0%

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